Department of Physics, IIT Madras

PH1020 Physics II 2018-19 | **Quiz I** | **21.02.2019**

Time: 8:00 am to 8:50 am | *Answer all questions* | **Max. marks: 20**

Name	Roll No.	Old Roll No.(if any)

Instructions: You must write the answers only in the allotted box. All vectors must be indicated clearly using arrows. You may use the reverse side of all pages for rough work. All symbols have their usual meaning unless stated otherwise. All constants are of appropriate dimensions. Further assume that unless otherwise stated all constants are positive. You must use only black or blue ink for writing the answers. Pencil must not be used even in sketches. Calculators, cell phones or any internet connectable device must not be in your possession during the examination.

Exam Hall No.

Total Marks

Q1	Q2	Q3	Q4	Q5	Q6

Use following formula for divergence and curl in spherical polar coordinate

$$\vec{\nabla}\cdot\vec{A}=\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2A_r\right)+\frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}\left(A_\theta\sin\theta\right)+\frac{1}{r\sin\theta}\frac{\partial A_\phi}{\partial\phi}$$

$$\vec{\nabla}\times\vec{A}=\frac{\hat{r}}{r\sin\theta}\left[\frac{\partial}{\partial\theta}\left(A_\phi\sin\theta\right)-\frac{\partial A_\theta}{\partial\phi}\right]+\frac{\hat{\theta}}{r\sin\theta}\left[\frac{\partial A_r}{\partial\phi}-\sin\theta\frac{\partial}{\partial r}\left(rA_\phi\right)\right]+\frac{\hat{\phi}}{r}\left[\frac{\partial}{\partial r}\left(rA_\theta\right)-\frac{\partial A_r}{\partial\theta}\right]$$

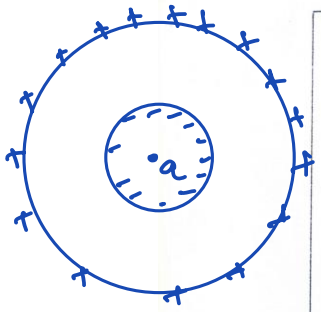
Use following formula for divergence and curl in cylindrical coordinate

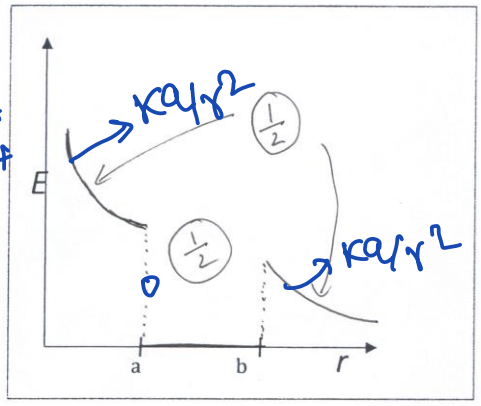
$$\vec{\nabla}\cdot\vec{A}=\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(A_\rho\rho\right)+\frac{1}{\rho}\frac{\partial A_\phi}{\partial\phi}+\frac{\partial A_z}{\partial z}$$

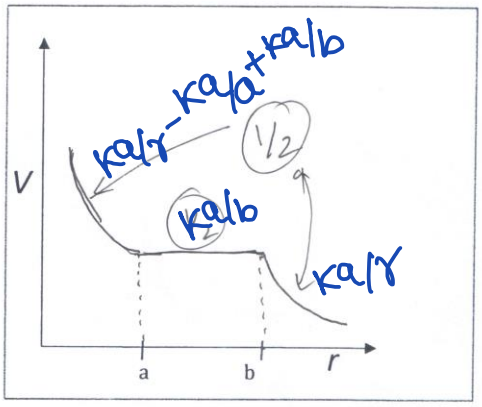
$$\vec{\nabla}\times\vec{A}=\hat{\rho}\left(\frac{1}{\rho}\frac{\partial A_z}{\partial\phi}-\frac{\partial A_\phi}{\partial z}\right)+\hat{\phi}\left(\frac{\partial A_\rho}{\partial z}-\frac{\partial A_z}{\partial\rho}\right)+\hat{z}\left(\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(A_\phi\rho\right)-\frac{1}{\rho}\frac{\partial A_\rho}{\partial\phi}\right)$$

1. In the boxes provided clearly indicate your answers to each of the questions below. The answers to each of these questions **must be a number or an expression or sketch.** Each box carries 1 **mark.**

(a) A positive charge q is placed at the centre of a thick spherical conducting shell of inner radius a and outer radius b. Sketch magnitude of electric field (E) and potential (V) as function of r (distance from the centre). (2 marks)







(b) Polarization within a sphere of radius R is given by $\vec{P} = \frac{P_0}{r}\hat{e}_r$. Where P_0 is a constant of appropriate dimensions (3 marks)

i. The bound surface charge density on the surface of the sphere is:

P_0/R

ii. The bound volume charge density in the sphere is:

$-P_0/r^2$

iii. The electric field inside the sphere is :

$E_r = -\frac{P_0}{\epsilon_0 r}$ or $\vec{E} = -\frac{P_0}{\epsilon_0 r}\hat{e}_r$

NO VECTOR SIGN THE (1/2)

2. Indicate the correct alternative(s) in the box provided. No marks will be awarded if any incorrect answer is given. [2 marks]

(A) Work done over a closed path in electrostatics is zero.

(B) A charge kept outside a conducting shell will have its presence felt at the centre of the shell.

(C) Tangential component of electric field is continuous at the interface of two media having surface charge density σ.

(D) For a point charge placed away from the origin, all moments in the multipole expansion except monopole moment identically vanish.

A, C

GIVE 1 MARK FOR ONE CORRECT CHOICE
0 FOR ANY WRONG CHOICE

3. In the box provided, indicate whether the following statements are **True** or **False**.
[4 × 1 = 4 marks]

(a) The function $\vec{F}(\rho, \varphi, z) = {}^{a}/_{\rho^2}\,(cos\varphi\ \hat{e}_\rho + sin\varphi\hat{e}_\varphi\ )$ represents an electrostatic field. True

$\frac{d}{d\rho}(\rho \vec{F}_\varphi) = \frac{d}{d\varphi}(F_\rho)$

$-sin\varphi/\rho^2 = -sin\varphi/\rho^2$

(b) Let $\vec{E}_1(r)$ and $\vec{E}_2(r)$ be the electrostatic fields produced by charge distributions $\rho_1(\mathbf{r})$ and $\rho_2(\mathbf{r})$ respectively. Then, the electrostatic energy due to the charge distribution $(\rho_1 + \rho_2)$ is given by $\frac{\epsilon_0}{2}\int(E_1^2 + E_2^2 + 2E_1 \cdot E_2)d\tau$. True

(c) The tangential component of $\vec{D}$ is always continuous across an interface separating two dielectric media which have different dielectric constants. False

(d) For a point charge $+Q$ placed inside a conducting shell, the electrostatic field at any point outside the conducting shell is zero. false

4. An infinitely long thin line charge lies along the z-axis with a uniform charge density λ. An infinitely long annular cylinder of inner radius a and outer radius b with its axis coinciding with z-axis has a volume charge density $\rho_c(\rho, \varphi, z) = k\rho$. Where k is constant of appropriate dimensions. Calculate electric field for all values of $\rho > 0$. (3 Marks)

line charge: $E \times 2\pi r \ell = \frac{\lambda \ell}{\varepsilon_0} \Rightarrow E = \frac{\lambda}{2\pi\varepsilon_0 r}$

$\rho < a$: $E = \frac{\lambda}{2\pi\varepsilon_0 r}$

$a < \rho < b$: $E = \frac{\lambda}{2\pi\varepsilon_0 r} + \frac{k(\rho^3 - a^3)}{3\varepsilon_0 r}$

$\rho > b$: $E = \frac{\lambda}{2\pi\varepsilon_0 r} + \frac{k(b^3 - a^3)}{3\varepsilon_0 r}$

$E \times 2\pi r \ell = \frac{\int_a^r k r \times 2\pi r dr \ell}{\varepsilon_0}$

$= \frac{2\pi k \ell (r^3 - a^3)}{3\varepsilon_0}$

6. Charge density in one dimension in a region $0 < x < a$ is given by $\rho = \rho_0 \frac{x}{a}$. Calculate the potential and the electric field in the region $0 < x < a$. The boundary conditions that has to be satisfied are $E = 0$ at $x = 0$ and $V = 0$ at $x = a$. (3 Marks)

$\nabla^2 \phi = \frac{\rho_0 x}{a\varepsilon_0}$

$\Rightarrow \frac{d^2\phi}{dx^2} = -\frac{\rho_0 x}{a\varepsilon_0} \Rightarrow \frac{d\phi}{dx} = -\frac{\rho_0 x^2}{2a\varepsilon_0} + C_1$

$E = \frac{\rho_0 x^2}{2a\varepsilon_0} - C_1$

$C_1 = 0$

$\Rightarrow \phi = -\frac{\rho_0 x^3}{6a\varepsilon_0} + C_1 x + C_2$

$C_2 = \frac{\rho_0 a^2}{6a\varepsilon_0}$

$\phi = \frac{\rho_0 (a^3 - x^3)}{6a\varepsilon_0}$ $E = \frac{\rho_0 x^2}{2a\varepsilon_0}$

5. A charge configuration in Cartesian coordinates is given as

$$\rho(x') = q\,\delta(x' + a) + q\,\delta(x' - a) - 2q\,\delta(x')$$

Find the monopole, dipole and quadrupole moments of the charge distribution given above. (3 Marks)

Monopole: $q = \int \rho(x')\,dx' = q + q - 2q = 0$

Dipole: $p = \int x' \rho(x')\,dx' = -qa + qa = 0$

Quadrupole:

$$Q_{xx} = \int \rho(x')\,dx'\,(3x_i^2 - r_i^2)$$

$$= 2\int x_i^2\,\rho(x')\,dx' = 2\left(q(-a)^2 + q(a)^2\right)$$

$$= 4qa^2$$

$$Q_{yy} = Q_{zz} = -2qa^2$$

$$Q_{xy} = \int \rho(x')\,dx'\,(3x_i y_i) = 0 = Q_{ij} \quad (i \neq j)$$